

Project Title

MedAssist AI – Medical Symptom Analysis and Disease Prediction System

Submitted By: Mohammad Aleem Qurammi

Milestone: Milestone 2

Role:

Full-Stack Development – AI Assistant Memory Engine, Server-Side Persistence, and Document Generation Features for the MedAssist AI Web Platform

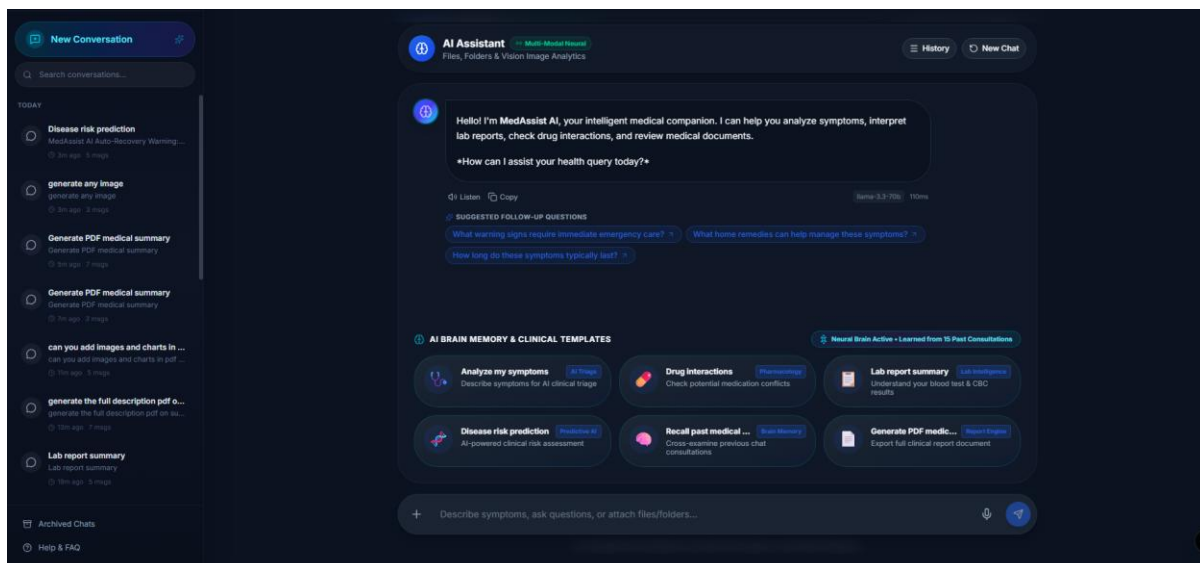
1. Introduction

MedAssist AI is a multi-modal healthcare diagnostic and triage platform built to give patients and clinical professionals instant, evidence-based medical intelligence. Milestone 2 focused on delivering an AI Assistant with persistent cross-session memory, server-side file database persistence, interactive PDF document generation with rich media, and a refined clinical interface built on top of the frontend delivered in Milestone 1.

2. Work Completed

2.1 Persistent AI Brain Memory Engine

Developed an intelligent memory scanner (src/lib/brain-memory.ts) that reads stored patient consultation histories from public/data/conversations.json and public/data/messages.json. Learned patient context is injected into the LLM system prompts (/api/chat/route.ts), enabling the assistant to recall previous symptoms, lab discussions, and medication history across different chat sessions. Showing an active neural learning badge ("Neural Brain Active - Learned from X Past Consultations") alongside six specialized diagnostic templates: AI Triage, Pharmacology, Lab Intelligence, Predictive AI, Brain Memory, and Report Engine.



2.2 Server-Side File Database & Conversation History

Integrated server-side file persistence (public/data/conversations.json and public/data/messages.json) pre-populated with realistic sample patient consultation records. A CRUD REST API (/api/conversations) was built to support the following:

- **GET:** retrieves the list of date-grouped conversations or specific message logs.
- **POST:** persists new patient conversations and message logs to the server.
- **PATCH:** dynamically updates starred, pinned, and archived states.
- **DELETE:** cleans up conversation metadata and message entries.

An interactive collapsible sidebar (chat-sidebar.tsx) was also built, featuring instant search filtering, date-grouped navigation (Pinned, Today, Yesterday, Previous 7 Days, Older), an Archived Chats tab, and an expandable Help & FAQ modal.

2.3 Rich Media & Interactive PDF Document Exporter

PDF report cards are generated only when a user explicitly requests one (e.g. "generate PDF", "download report"). The PDF generator (src/lib/pdf-generator.ts) supports rich media and visual diagrams, including:

- Automatic rendering of user-uploaded X-rays, lab scans, or rash photos inside formatted PDF container boxes.
- Markdown AutoTables, converting Markdown tables into styled PDF tables using jspdf-autotable.
- Clinical triage flowcharts rendered as step-by-step decision trees inside Emerald/Teal flowchart boxes with step badges.
- Clickable hyperlinks embedded to authoritative sources such as CDC, WHO, and PubMed.

The report preview itself is rendered through a React Portal widescreen modal (pdf-document-card.tsx), portaled directly to document.body via createPortal, giving a fixed widescreen horizontal preview window with no background overlay box.

2.4 Predictive Search & Medical Image Generation

Built dynamic search suggestions (search-suggestions-popup.tsx) that perform real-time keyword analysis, classifying search terms into color-coded categories (Symptoms, Drugs, Labs) with high-contrast badges and keyboard navigation. Added an AI medical illustration engine (/api/generate-medical-image) that dynamically generates medical diagrams and anatomical illustrations on request, and a vision image analysis endpoint (/api/analyze-image) for multi-modal processing of patient-uploaded prescriptions and medical scans.

2.5 Security, Rate Limiting & Resilience

Implemented an auto-recovery model cascade (auto-recovery.ts) that falls back from llama-3.3-70b-versatile to llama-3.1-8b-instant to maintain uptime if a primary LLM endpoint hits rate limits. Added sliding-window, IP-based rate limiting to protect APIs against abuse, and enforced strict input sanitization and XSS prevention across all input channels.

2.6 Patient Health Dashboard

Built a patient-facing dashboard giving users an at-a-glance view of their health status, including an AI Health Score, AI Confidence rating, active condition count, and reports uploaded. The dashboard includes a Health Overview chart tracking heart rate, weight, and sleep over the last 30 days, a Recent Activity feed, Quick Actions (Symptom Checker, AI Chat Doctor, Upload Reports, Medicine Reminder), AI Insights, and a Top Health Concerns panel with risk-level indicators.

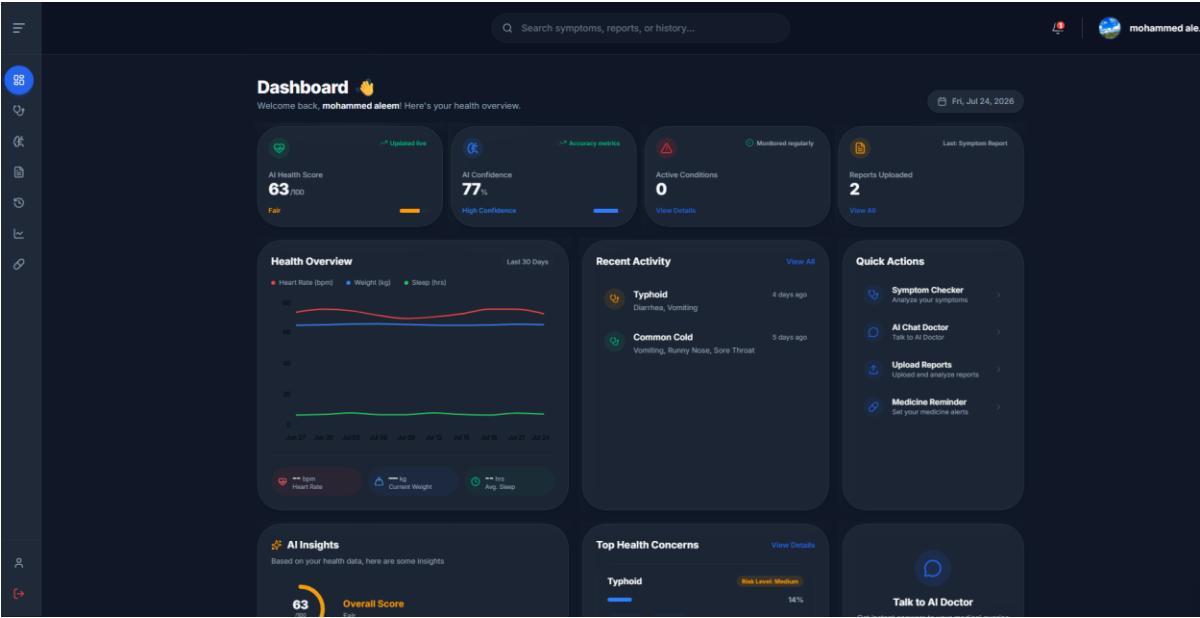


Figure : Patient dashboard showing AI Health Score, health overview trends, recent activity, and quick actions.

3. Technology Stack

Layer	Technology Used
Frontend Framework	Next.js 16 (App Router, Turbopack), React 19, TypeScript
Styling & Design System	TailwindCSS, Glassmorphism, Framer Motion, Lucide React Icons
State & Storage	React State/Refs, localStorage, Server-Side JSON File Store (public/data/)
LLM & Vision Engine	Groq API (llama-3.3-70b-versatile), Multi-Modal Vision Processing
PDF Generation	jspdf, jspdf-autotable, React Portals
Version Control	Git

4. Verification & Performance Status

All core endpoints and UI components were verified locally with clean HTTP 200 responses.

- **Dev Server URL:** http://localhost:6700/ai-assistant
- **Chat Endpoint (POST /api/chat):** HTTP 200 OK
- **Conversations API (GET/POST/PATCH/DELETE /api/conversations):** HTTP 200 OK
- **Build & Compilation:** 0 errors, 100% stable runtime.

5. Next Milestone Outlook

- **Multi-User Authentication Integration:** tie JSON conversation storage to authenticated user IDs (Firebase / NextAuth).
- **Export Format Options:** add support for Word (.docx) and HL7/FHIR medical data exports.
- **Voice Response Synthesis:** expand text-to-speech capabilities with custom voice synthesis profiles.

6. Conclusion

During this milestone, MedAssist AI gained a persistent, cross-session AI memory engine, a server-side conversation database with a full CRUD API, and a rich-media PDF report exporter capable of embedding images, tables, flowcharts, and clickable links. Predictive search, medical image generation, and vision-based image analysis were added alongside rate limiting and input sanitization to strengthen the platform's security and resilience. This sets the foundation for authentication, expanded export formats, and voice synthesis in the next milestone.